(4) A gene g is a marker for target group t against the groups in C if and only if g is a pairwise marker for t against every group in C.

(5) If a study reports $M_{s,p}(t,h)$ for each pairwise target–comparison contrast in a partition, then a later user can reconstruct $M_{s,p}(t;C)$ for any set C of comparison groups contained in that partition by taking an intersection.

(6) A gene g is a marker for the marker-defined cell type claim I if and only if g is a marker for every scoped local group in I:

$$g \in M(I) \iff \forall i \in I,\ g \in M_{s_i,p_i}(t_i;C_i).$$

(7) If $C_{\text{local}} \subseteq C_{\text{global}}$, then

$$M_{s,p}(t;C_{\text{global}}) \subseteq M_{s,p}(t;C_{\text{local}}).$$

(8) If $C_{\text{state}} \subseteq C_{\text{type}}$, then

$$M_{s,p}(t;C_{\text{type}}) \subseteq M_{s,p}(t;C_{\text{state}}).$$

(9) Let C_{parent} be a broad comparison set and let $C_{\text{child}} \subseteq C_{\text{parent}}$ be the local comparison set induced by a refinement of that parent group. Then

$$M_{s,p}(t;C_{\text{parent}}) \subseteq M_{s,p}(t;C_{\text{child}}).$$

(10) If B is a background set and weights w_h satisfy $\sum_{h\in B} w_h = 1$, then

$$\mu(s,p,t,g) - \sum_{h\in B} w_h \mu(s,p,h,g) = \sum_{h\in B} w_h \left(\mu(s,p,t,g) - \mu(s,p,h,g)\right).$$

(2) If F satisfies the neutral-element, pairwise-agreement, and conjunctive-comparison properties above, then for every finite set C of groups being compared against the target,

$$F(C) = \bigcap_{h\in C} M(t,h).$$

Equivalently,

$$g \in F(C) \iff \forall h \in C,\ g \in M(t,h).$$

(3) Let $C \subseteq K_{s,p} \setminus \{t\}$ be the groups that target t must be distinguished from. We call C the comparison set. The marker set for target group t against C is

$$M_{s,p}(t;C) = \bigcap_{h\in C} M_{s,p}(t,h).$$

Thus

$$g \in M_{s,p}(t;C) \iff \forall h \in C,\ g \in M_{s,p}(t,h).$$

(12) Let k be the number of global cell types, let m be the number of local experiments, and suppose each experiment contributes at most q ordered distinct comparison pairs among those global cell types. If the experiments cover every ordered distinct global pair, then

$$k^2 - k \le mq.$$

(1) Fix a gene set G, a study, partition, target group t, and pairwise marker sets $M(t,h)$, where h is one group being compared against t. Let C be the set of groups that t must be distinguished from, and let $F(C)$ be the marker genes assigned to t for that task. We require:

1. Scopedness. A marker is tied to a target and to the groups the target is being distinguished from, not to a gene or target–gene pair alone.
2. Admissibility. A biological marker claim compares the target to at least one other group and does not compare the target to itself. The empty set is only the neutral base case: $F(\varnothing) = G$.
3. Pairwise agreement. If the target is compared to one group, the rule gives the usual pairwise marker set: $F(\{h\}) = M(t,h)$.
4. Conjunctive comparison. If the target is compared to two sets of groups, a marker must work for both tasks:

   $$F(C_1 \cup C_2) = F(C_1) \cap F(C_2).$$

5. Biological relevance. Adding groups makes the marker claim harder, not easier; it cannot rescue a gene that already failed a required contrast.

(11) Suppose a collection of local experiments covers every required pairwise comparison in a global comparison set, and suppose the same marker panel separates each covered local experiment. Then the marker panel separates the global set.

(13) Let P_i be the set of ordered distinct comparison pairs contributed by study i. Adding a new study contributes only the pairs not already observed:

$$|P_{\text{old}} \cup P_{\text{new}}| = |P_{\text{old}}| + |P_{\text{new}}| - |P_{\text{old}} \cap P_{\text{new}}|.$$

For two studies observing r_1 and r_2 relevant cell types and sharing s of them, the unordered version is

$$\binom{r_1}{2} + \binom{r_2}{2} - \binom{s}{2}.$$

(14) Flattening a scoped marker claim to a cell type–gene pair,

$$(t,C,g) \mapsto (t,g).$$

is not injective in the comparison set.

(15) If a panel S separates all profiles in a finite universe K, then

$$|K| \le 2^{|S|}.$$

legend
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theorem
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